

Class06 Homework

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Q6. How would you generalize the original code above to work with any set of input protein structures?

Original Code:

```
# Can you improve this analysis code?  
library(bio3d)  
s1 <- read.pdb("4AKE") # kinase with drug
```

Note: Accessing on-line PDB file

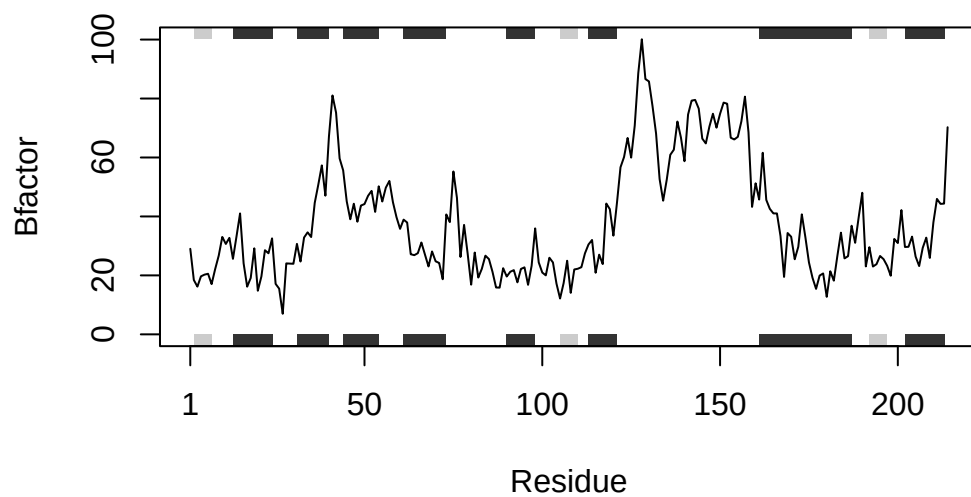
```
s2 <- read.pdb("1AKE") # kinase no drug
```

Note: Accessing on-line PDB file
PDB has ALT records, taking A only, rm.alt=TRUE

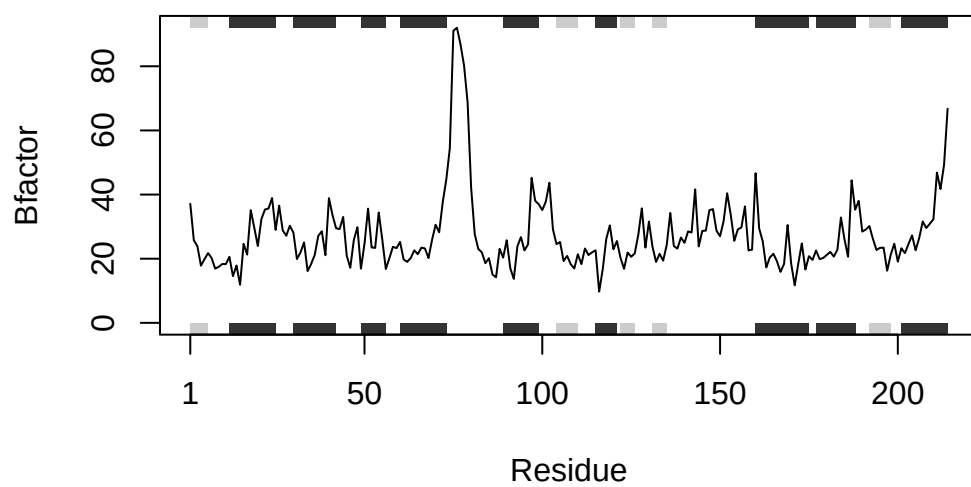
```
s3 <- read.pdb("1E4Y") # kinase with drug
```

Note: Accessing on-line PDB file

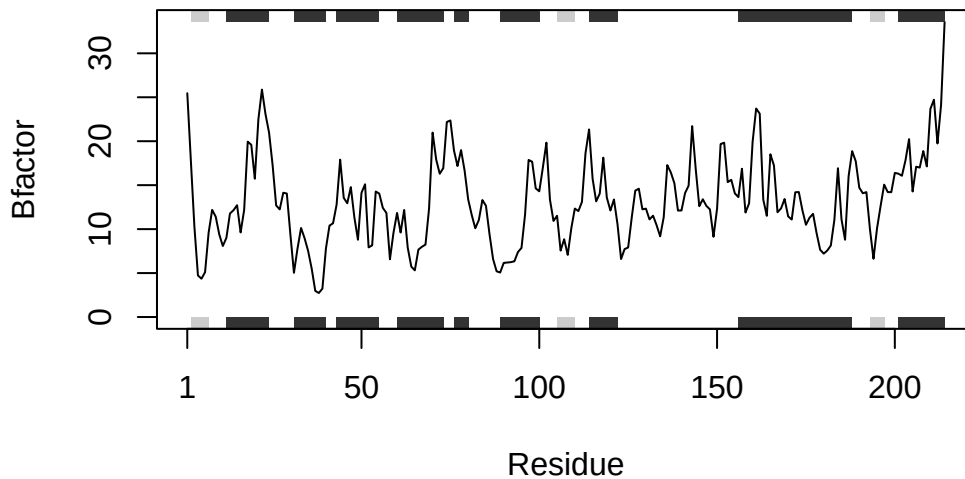
```
s1.chainA <- trim.pdb(s1, chain="A", elety="CA")  
s2.chainA <- trim.pdb(s2, chain="A", elety="CA")  
s3.chainA <- trim.pdb(s3, chain="A", elety="CA")  
s1.b <- s1.chainA$atom$b  
s2.b <- s2.chainA$atom$b  
s3.b <- s3.chainA$atom$b  
plotb3(s1.b, sse=s1.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s2.b, sse=s2.chainA, typ="l", ylab="Bfactor")
```



```
plotb3(s3.b, sse=s3.chainA, typ="l", ylab="Bfactor")
```



Generalized Code:

```
library(bio3d)

#Input: pdb_ids: a character vector of PDB IDs.
#chain: the protein chain to analyze, default is "A".

#What the function below does: reads each PDB structure,
#only keeps C-alpha atoms from the chosen chain,
#extracts B-factors, plots the B-factor profile for each structure

#output: returns a list containing the trimmed PDB objects,
#the extracted B-factor vectors, B-factor plots

analyze_pdb_bfactors <- function(pdb_ids, chain = "A") {

  # read and trim each structure
  trimmed_pdbs <- lapply(pdb_ids, function(id) {
    pdb <- read.pdb(id)
    trim.pdb(pdb, chain = chain, elety = "CA")
```

```

})

# extract B-factors
b_factors <- lapply(trimmed_pdbs, function(pdb) {
  pdb$atom$b
})

# plot each structure
for (i in seq_along(trimmed_pdbs)) {
  plotb3(
    b_factors[[i]],
    sse = trimmed_pdbs[[i]],
    typ = "l",
    ylab = "B-factor",
    main = paste("B-factor profile:", pdb_ids[i], "Chain", chain)
  )
}

# return results
return(list(
  pdb_ids = pdb_ids,
  chain = chain,
  trimmed_pdbs = trimmed_pdbs,
  b_factors = b_factors
))
}

```

Example:

```
result <- analyze_pdb_bfactors(c("4AKE", "1AKE", "1E4Y"), chain = "A")
```

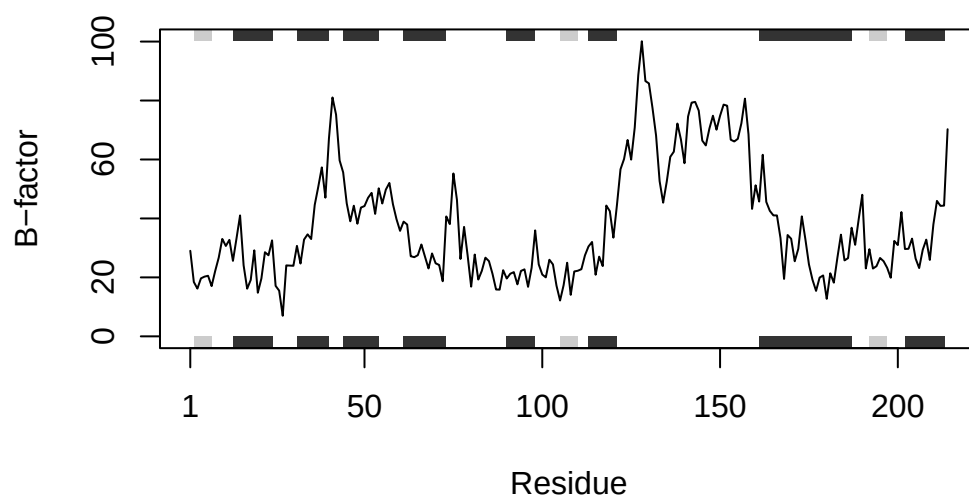
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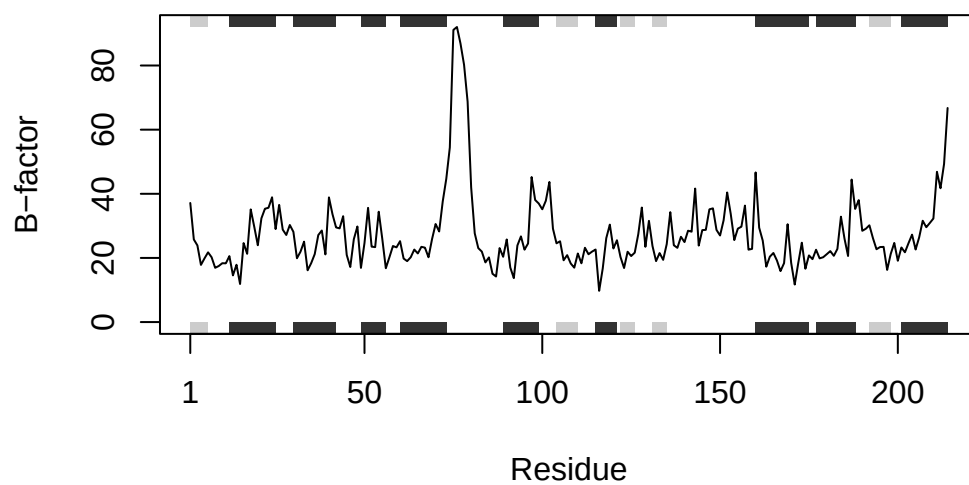
PDB has ALT records, taking A only, rm.alt=TRUE

Note: Accessing on-line PDB file

B-factor profile: 4AKE Chain A



B-factor profile: 1AKE Chain A



B-factor profile: 1E4Y Chain A

